

Final Project Report

Ziwei Zhang
zz4980

zz4980@nyu.edu

Abstract

Access to public restrooms in urban environments like New York City is a significant challenge for residents and visitors alike. This project addresses the critical issue of the availability and accessibility of public restrooms, specifically examining their spatial relationship with the city's extensive subway network. Using geospatial analysis techniques, this study analyzes public restroom distribution across NYC neighborhoods and measures their proximity to subway stations using an 800-meter buffer zone (approximately a 10-minute walk). Findings reveal significant disparities in restroom availability across different districts, with only 25% of subway stations having at least one restroom within walking distance. The analysis also highlights areas with concentrations of fully accessible restrooms versus areas lacking such facilities. These insights have important implications for urban planning, public health policy, and transportation accessibility, particularly for vulnerable populations including those with disabilities, the elderly, and families with young children.

1. Introduction

Having reliable access to toilets and handwashing facilities is a basic human right (Swayne et al., 2023). This issue is especially critical in dense urban settings, where people and other species congregate and bacteria and disease can spread easily. Cities like New York face increasing challenges in meeting public sanitation demand. Individuals with heightened mobility or housing precarity are more likely to experience difficulties finding a clean, safe, and accessible facility.

Despite sanitation being a fundamental environmental determinant of health, public restroom availability in the U.S. has been declining for decades, resulting in serious consequences for public health and equity (Brown, 2020). People experiencing homelessness, for example, are frequently forced to relieve themselves in undesignated places, contributing to risks of disease transmission and criminalization. Other vulnerable groups, including women, people



Figure 1. A Public Bathroom run by NYC Parks

ple who menstruate, people with disabilities, and delivery workers, encounter daily barriers in accessing facilities that meet their basic needs with dignity.

Inspired by growing body of research, this project focuses on mapping and analyzing public restroom infrastructure in New York City through a lens of equity and accessibility. By combining spatial data with information on subway access and restroom amenities, I aim to develop an interactive tool that helps users locate restrooms based on proximity and accessibility features. This tool is especially relevant for populations like delivery drivers and unhoused individuals, who face compounded challenges navigating NYC's fragmented sanitation landscape. Ultimately, the goal is to support data-informed advocacy and urban planning that centers sanitation justice as a core pillar of public infrastructure.

2. Results and Demonstration

Figure 1 displays the distribution of bathrooms across New York City neighborhoods. Darker blue areas indicate neighborhoods with more bathrooms (7-17 per neighborhood), while lighter shades and white areas have fewer bathrooms (0-2). Manhattan, particularly in midtown and downtown areas, shows the highest concentration of public bathrooms, with several dark blue hotspots. Outer boroughs generally have fewer facilities, with Staten Island and parts of Queens showing the lowest bathroom availability.

Figure 2 ranks the subway stations with the highest number of bathroom facilities. The 5 Ave/53 St station leads

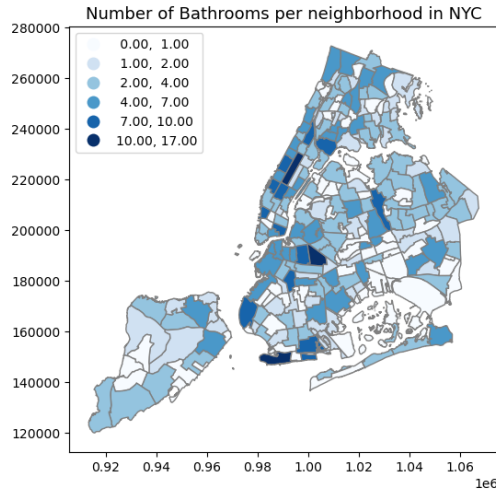


Figure 2. Number of Bathrooms per neighborhood in NYC

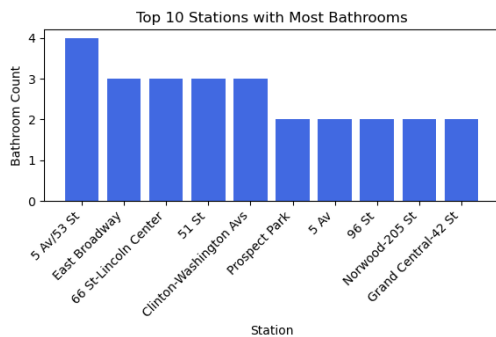


Figure 3. Top 10 Stations with Most Bathrooms

significantly with 4 bathrooms, while East Broadway, 66 St-Lincoln Center, 51 St, and Clinton-Washington Aves follow with 3 bathrooms each. The remaining stations in the top 10 (Prospect Park, 5 Av, 96 St, Norwood-205 St, and Grand Central-42 St) each have 2 bathrooms available. This highlights where travelers can most reliably find facilities in the transit system.

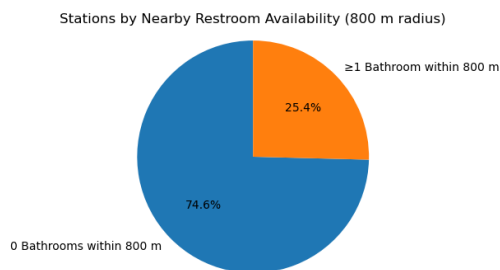


Figure 4. Stations by Nearby Restroom Availability (800 m radius)

Figure 4 reveals a significant accessibility issue in the

NYC transit system: only 25.4% of stations have at least one bathroom within an 800-meter radius (about a 10-minute walk). The overwhelming majority (74.6%) have zero bathrooms within this reasonable walking distance. This visualization emphasizes the bathroom desert problem facing many commuters and tourists using public transportation in NYC.

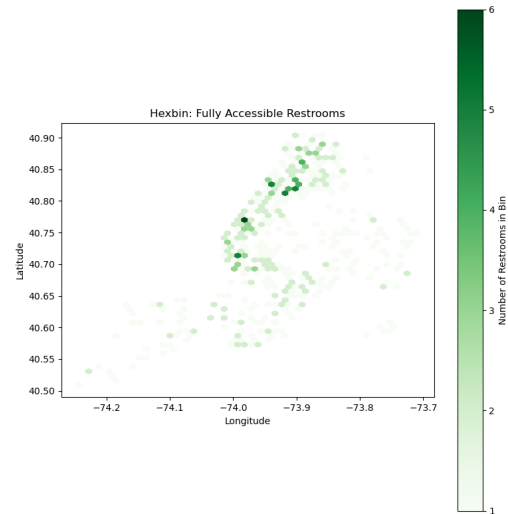


Figure 5. Hexbin Heatmap: Fully Accessible Restrooms)

Figure 5 maps the distribution of fully accessible restrooms across NYC using a hexagonal grid system. The color intensity (from light to dark green) indicates the concentration of accessible facilities in each area, with darker green representing 5-6 accessible restrooms. The map shows clusters of accessibility in specific areas (midtown and parts of upper Manhattan), while much of the city has minimal accessible bathroom options, appearing as very light green or white hexagons.

3. Implementation

This project combined geospatial analysis and data visualization to examine the relationship between public restrooms and subway stations in NYC. Two core datasets were used: the NYC Public Restrooms dataset and the MTA Subway Stations dataset (both dated April 10, 2025), each containing geographic coordinates, accessibility attributes, and operational details.

Preprocessing steps included filtering restroom records for completeness and operational status, converting both datasets to GeoDataFrames using GeoPandas, and reprojecting them to the New York State Plane coordinate system (EPSG:2263) to enable accurate spatial measurements.

The spatial analysis comprised several components. Initial data exploration used Folium to visualize the distribu-

tion of facilities and stations. Spatial joins with Community District boundaries were performed to quantify restroom and station density by district. An 800-meter buffer analysis assessed restroom proximity to subway stations, while further analysis evaluated the spatial distribution of fully accessible facilities.

Various visualization techniques were employed: choropleth maps to show facility density, buffer maps to highlight walking-distance access, hexbin maps to detect clusters of accessible restrooms, and bar/pie charts to summarize station-level access. All analyses were conducted in Python using GeoPandas, Matplotlib, Seaborn, and Folium, with careful attention to coordinate reference systems for spatial accuracy.

4. Discussion

The analysis revealed pronounced disparities in restroom access across NYC. While some districts are well served, others have limited or no restroom infrastructure. Only 25% of subway stations have at least one restroom within an 800-meter walk, indicating a significant gap in essential amenities for commuters and visitors. Accessible facilities are heavily clustered in Midtown and Uptown Manhattan, while large areas of the outer boroughs remain underserved.

These findings emphasize the need for integrated planning between transit infrastructure and public amenities. The MTA and city agencies should prioritize investments in transit zones that currently lack adequate restroom access. Addressing accessibility disparities is particularly important for seniors, individuals with disabilities, and families with young children.

This study also has limitations. It did not fully account for restroom operating hours due to inconsistencies in the source data's text fields. Additionally, the dataset may exclude semi-public facilities such as restrooms in restaurants or cafes. Factors such as restroom quality, cleanliness, and walkability barriers were also not assessed. The use of a fixed 800-meter buffer assumes ideal walking conditions, without accounting for real-world obstacles like highways or inaccessible infrastructure.

References

1. **Developing Evidence for Building Sanitation Justice: A Multi-Methods Approach to Documenting Public Restroom Inequities in San Francisco** *International Journal of Environmental Research and Public Health*, 20(12), 6058. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10343041/>
2. **The pandemic has closed public restrooms, and many have nowhere to go.** *Stateline*. <https://stateline.org/2020/07/23/the-pandemic-has-closed-public-restrooms-and-many-have-nowhere-to-go/>

[have-nowhere-to-go/](https://stateline.org/2020/07/23/the-pandemic-has-closed-public-restrooms-and-many-have-nowhere-to-go/)

3. **On-Street Toilets for Sanitation Access in Urban Public Spaces: A Review** *Cities*, 108, 102974. <https://www.sciencedirect.com/science/article/abs/pii/S0957178721000205>

4. **Does Visualization Matter? The Role of Interactive Data Visualization to Make Sense of Information** *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13. https://www.researchgate.net/publication/325025606_Does_Visualization_Matter_The_Role_of_Interactive_Data_Visualization_to_Make_Sense_of_Information

5. **Figure 1, Public Restroom Facility at NYC Park** *NYC Department of Parks and Recreation*. <https://www.nycgovparks.org/facilities/restrooms>